

String Instructions (8086 Microprocessor)

What is a String?

A **string** is a sequence of **bytes or words** stored in **successive memory locations**. String instructions are special 8086 instructions that automatically:

- Access memory using index registers
- Update addresses without extra instructions
- Process arrays efficiently

1. Registers Used in String Instructions

- **SI (Source Index)** → points to source string (DS:SI)
- **DI (Destination Index)** → points to destination string (ES:DI)
- **CX** → repetition counter (used by REP)
- **DF (Direction Flag)** → controls address movement

Direction Flag (DF)

- DF = 0 → auto **increment** (forward processing)
- DF = 1 → auto **decrement** (reverse processing)

Control instructions:

- **CLD** → Clear DF
- **STD** → Set DF

Exam Note

By default, source string uses **DS:SI** and destination string uses **ES:DI**.

2. LODS — Load String

Purpose

Loads one element of a string from memory into the accumulator.

$$AL/AX \leftarrow [DS : SI]$$

After execution:

- SI is automatically updated
- Increment or decrement depends on DF

Variants

- LODSB $\rightarrow$ byte $\rightarrow$ AL (SI $\pm$ 1)
- LODSW $\rightarrow$ word $\rightarrow$ AX (SI $\pm$ 2)

Listing 1: LODS Example

```

1 CLD
2 MOV SI, OFFSET ARRAY
3 LODSB          ; AL <- ARRAY[0], SI++

```

Important

REP prefix **cannot** be used with LODS.

3. STOS — Store String

Purpose

Stores accumulator content into destination string memory.

$$[ES : DI] \leftarrow AL/AX$$

- STOSB $\rightarrow$ store byte (DI $\pm$ 1)
- STOSW $\rightarrow$ store word (DI $\pm$ 2)

Listing 2: STOS Example

```

1 MOV AL, 5AH
2 CLD
3 STOSB          ; ES:DI <- AL

```

4. MOVS — Move String

Purpose

Copies data from source string to destination string.

$$[ES : DI] \leftarrow [DS : SI]$$

Both SI and DI are automatically updated.

- MOVS_B → move byte
- MOVS_W → move word

Listing 3: MOVS Example

```
1 CLD
2 MOV CX, 10
3 REP MOVSB ; copy 10 bytes
```

5. SCAS — Scan String

Purpose

Searches a string for a specific value.

- SCAS_B → compare AL with ES:DI
- SCAS_W → compare AX with ES:DI

Comparison affects **Zero Flag (ZF)**.

Listing 4: SCAS Example

```
1 MOV CX, 100
2 MOV AL, 0AH
3 CLD
4 REPNE SCASB ; search until match or CX=0
```

6. INS and OUTS Instructions

INS

Transfers data from I/O port to memory.

$$ES : DI \leftarrow [DX]$$

- INSB, INSW, INSD

OUTS

Transfers data from memory to I/O port.

$$[DX] \leftarrow DS : SI$$

- OUTSB, OUTSW, OUTSD

7. REP Prefix

REP (Repeat)

Automatically repeats a string instruction using CX as counter.

- CX decrements after each execution
- Stops when $CX = 0$
- REP MOVSB, REP STOS
- REPE / REPZ $\rightarrow$ repeat while $ZF = 1$
- REPNE / REPNZ $\rightarrow$ repeat while $ZF = 0$

8. Quick Summary

One-Page Recap

- LODS $\rightarrow$ memory $\rightarrow$ accumulator
- STOS $\rightarrow$ accumulator $\rightarrow$ memory
- MOVS $\rightarrow$ memory $\rightarrow$ memory
- SCAS $\rightarrow$ search in string
- REP $\rightarrow$ automatic repetition
- DF controls direction